

Recitation 14 — Answer Key

Loanable Funds, Savings, and Bond Pricing

Question 2 — Google's Project Decisions and Demand for Loanable Funds

The data on Google's five projects:

Project	Cost	Break-even Interest Rate
A	\$100 million	8%
B	\$50 million	12%
C	\$200 million	5%
D	\$25 million	4%
E	\$150 million	10%

Decision rule: Take on a project only if the market interest rate is *strictly less than* the project's break-even rate (otherwise borrowing costs swallow the project's returns).

(a) Which projects at 11% and at 6%?

At an interest rate of 11%: Only Project B (break-even 12%) clears the bar.

- Project B: $12\% > 11\%$ ✓ **Undertake**
- Projects A, C, D, E: break-even rates (8%, 5%, 4%, 10%) are all below 11% — skip.

Answer: At 11%, Google undertakes **Project B only**, borrowing **\$50 million**.

At an interest rate of 6%: All projects with break-even rates above 6% are profitable.

- Project B (12%), Project E (10%), Project A (8%): all above 6% ✓
- Project C (5%) and Project D (4%): below 6% — skip.

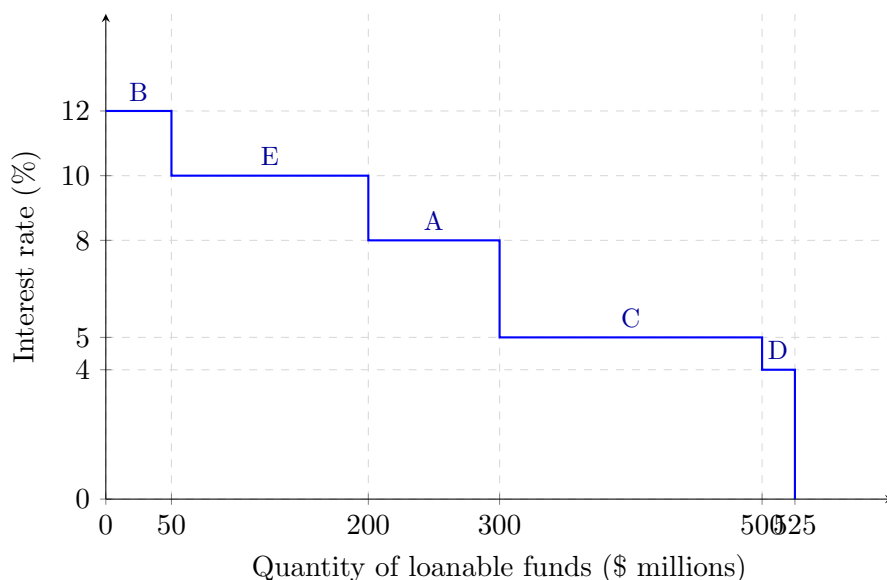
Answer: At 6%, Google undertakes Projects **A, B, and E**, borrowing $100 + 50 + 150 = \mathbf{\$300}$ million.

(b) Demand curve for loanable funds

Rank projects from highest break-even rate (taken first) to lowest, and accumulate the borrowed amount:

Project	Break-even Rate	Cost	Cumulative Quantity Demanded
B	12%	\$50M	\$50M
E	10%	\$150M	\$200M
A	8%	\$100M	\$300M
C	5%	\$200M	\$500M
D	4%	\$25M	\$525M

The demand curve is a downward-sloping step function:



Answer: Each horizontal segment corresponds to one project; the vertical drops occur at the cumulative \$ amount where Google moves on to the next, lower-break-even project.

Question 3 — Shifts in the Supply of Savings

(c) Repeal of automatic enrollment in retirement plans

Auto-enrollment exploits inertia: once enrolled, most workers never opt out, so 5% of their pay flows to savings by default. If Congress abolishes the law, fewer workers would actively choose to enroll, and aggregate savings at every interest rate would fall.

Answer: The supply of savings shifts **leftward** (curve **a**). At any given interest rate, the quantity of savings is smaller.

(d) “Carpe diem” / “take no thought for the morrow”

All three cultural messages encourage present-oriented consumption rather than future-oriented saving. If Americans collectively become more present-biased, they save less at every interest rate.

Answer: The supply of savings also shifts **leftward** (curve **b**).

Question 3 (Bonds) — Zero-Coupon Bond Yields

A zero-coupon bond pays the face value F at maturity and nothing in between. With one year to maturity, if the price today is P , the implicit one-year interest rate i satisfies

$$P = \frac{F}{1+i} \iff i = \frac{F-P}{P}.$$

(e) **Bond A:** $F = \$1,000$, $P = \$925$

$$i_A = \frac{1000 - 925}{925} = \frac{75}{925} \approx 0.0811 = 8.11\%.$$

Answer: The interest rate on Bond A is approximately **8.11%**.

(f) **Bond B:** $F = \$1,000$, $P = \$900$

$$i_B = \frac{1000 - 900}{900} = \frac{100}{900} \approx 0.1111 = 11.11\%.$$

Answer: The interest rate on Bond B is approximately **11.11%**.

(g) **Which bond would you rather invest in? Think again!**

First instinct: Bond B looks better — a higher yield (11.11% vs. 8.11%) on the same face value and same maturity.

Think again: If two otherwise-identical bonds trade at noticeably different prices, the market is telling us something. The cheaper bond (Bond B) almost certainly carries higher **default risk** — the market discounts it more heavily because there is a real chance the issuer fails to pay the full \$1,000 at maturity. The extra ≈ 3 percentage points of yield is a **risk premium**, not a free lunch.

Answer: The “right” choice depends on your tolerance for risk.

- If both bonds are issued by the *same* borrower with identical terms, Bond B is unambiguously better (same risk, higher yield) — but in practice this would be an arbitrage opportunity that markets erase quickly.
- More realistically, Bond B is priced lower because investors view it as *riskier*. A risk-averse investor may rationally prefer Bond A despite the lower yield.

The lesson: **higher yields on otherwise-similar bonds usually reflect higher risk**, not a market inefficiency you can exploit.